Homework Assignment 4

Math 311, Fall 2026

Rutgers University

Exercise 2.5.1

(a) By Bolzano Wierstrass, every bounded sequence has a convergent subsequence. So if a sequence has a bounded subsequence, then that subsequence must also contain a convergent subsequence, which would be a subsequence of the main sequence as well.

(b) Define

$$x_{2n-1} = \frac{1}{n+1}, \qquad x_{2n} = 1 - \frac{1}{n+1}.$$

Then $x_n \neq 0, 1$ but

$$x_{2n-1} \to 0, \qquad x_{2n} \to 1.$$

(c) If we combine the sequences

$$(1), (1, \frac{1}{2}), (1, \frac{1}{2}, \frac{1}{3}), \dots$$

together to form

$$1, 1, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{3}, \dots$$

For each $n \in \mathbb{N}$, we can get a constant subsequence of $\frac{1}{n}$ by choosing the nth element of each "block".

(d) This is impossible, since the sequence formed by the infinite set ordered, eventually converges to 0, but 0 is not in the set.

Exercise 2.5.2

(a) True, consider the proper subsequence $(x_2, x_3, \dots)$, all the elements except the first. Then this subsequence converges by assumption. Adding x_1 to the front does not affect the tail, so (x_n) converges as well.

(b) True, if (x_n) converges, then every subsequence also converges. Thus the contrapositive means that if we a divergent subsequence, then the (x_n) diverges.

(c) True, by Bolzano Weierstrass, if (x_n) is bounded, then it must have a convergent subsequence, say $(x_{nk} \to L)$. Then since (x_n) is divergent, there exists $\epsilon > 0$ such that infinitely many terms satisfy $|x_n - L| \geq \epsilon > 0$. Using these terms, we can construct another bounded subsequence that never converges to L. Since this subsequence is bounded, we can reapply Bolzano Weierstrass and find a convergent sequence that converges to another limit that cannot be L since $\epsilon > 0$.

(d) Suppose (x_n) is increasing WLOG and contains a convergent subsequence $x_{nk} \to L$. Since the subsequence is convergent, it is also bounded by some upper limit say B. Then for every $n \in \mathbb{N}$, choose $nk \geq n$ so

$$x_n \leq x_{nk} < B.$$

Thus the (x_n) is also bounded above by B. Thus, by the monotone convergence theorem, $(x_n) \to L$.

Exercise 2.5.5 Assume (a_n) is bounded sequence where every convergent subsequence converge ot the same limit $a \in \mathbb{R}$. Since a_n is bounded, exercise 2.5.2c says that if a_n diverges, then it's subsequences converge to different limit. This contradicts our assumption that all convergent subsequences converge to a. Thus a_n is convergent. Then all its subsequences must all be convergent, and they all converge to a. Therefore $a_n \to a$.

Exercise 2.5.9 Let (a_n) be a bounded sequence, and define

$$S = \{x \in \mathbb{R} : x < a_n \text{ for infinitely many terms } a_n\}.$$

Since (a_n) is bounded, there exists some $B \in \mathbb{R}$ such that

$$-B \leq a_n \leq B \text{ for every } n \in \mathbb{N}.$$

Then $-B - 1 \in S$, so $S \neq \varnothing$ and every element $x \in S$ must be less than B since

$$x < a_n \leq B.$$

That is, S is bounded above, so by the axiom of completeness, S must have a least upper bound $s = \sup S$. We construct a subsequence (a_{nk}) out of S by recursively choosing increasing elements. Then a_{nk} is bounded as well, so by the monotone convergence theorem, a_{nk} converges to some limit L.

For each $k \in \mathbb{N}$, since $s - \frac{1}{k} < s$, it follows that $s - \frac{1}{k}$ is not an upper bound for S. So there exists $t \in S$ such that

$$s - \frac{1}{k} < t.$$

Then by definition of S, there are infinitely many terms that satisfy $a_n > t$. So infinitely many terms also satisfy

$$a_n > s - \frac{1}{k}.$$

While, only finitely many terms satisfy

$$a_n > s + \frac{1}{k}.$$

We can then recursively choose indices $n_1 < n_2 < \dots$ so that

$$s - \frac{1}{k} < a_{nk} \leq s + \frac{1}{k}.$$

That is,

$$|a_{nk} - s| \leq \frac{1}{k}.$$

Given $\epsilon > 0$, choose $K \in \mathbb{N}$ such that $\frac{1}{K} < \epsilon$. Then, for every $k \geq K$ we have

$$|a_{nk} - s| \leq \frac{1}{k} \leq \frac{1}{K} < \epsilon$$

Therefore, $a_{nk} \to s = \sup S$.

Exercise 2.6.2

(a) Let

$$x_n = \frac{(-1)^n}{n}.$$

Then $x_n \to 0$, so it is cauchy. However, it is not monotone since it alternates between positive and negative values.

(b) Impossible. Every cauchy sequence is convergent, every convergent sequence is bounded, and every subsequence of a bounded sequence is bounded.

(c) Impossibe. Since the subsequence is cauchy, it must also be convergent, and so it must also be bounded. Since the sequence is monotone, if it has a bounded subsequence, it must be bounded as well. Then the sequence is bounded and monotone, so by the monotoen convergent theorem, it must be convergent.

(d) Consider the sequence (x_n) defined by

$$x_{2n} = 0, \qquad x_{2n-1} = n.$$

Then the sequence is unbounded since $x_{2n-1} \to \infty$, but its even subsequence $x_{2n} \to 0$, and so x_n contains a cauchy.

Exercise 2.6.4

(a) Yes. Let $\epsilon > 0$. Since a_n and b_n are cauchy, choose N such that for all $m, n \geq N$,

$$|a_n - a_m| < \frac{\epsilon}{2}, \qquad |b_n - b_m| < \frac{\epsilon}{2}.$$

Then for every $m, n \geq N$ we have

$$\begin{aligned} |c_n - c_m| &= ||a_n - b_n| - |a_m - b_m|| \\ &\leq |(a_n - b_n) - (a_m - b_m)| \\ &= |(a_n - a_m) + (b_m - b_n)| \\ &\leq |a_n - a_m| + |b_n - b_m| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

Thus c_n is cauchy.

(b) Not cauchy. Consider $(a_n) = 1$. Then $a_n \to 1$ and so is cachy. But $c_n = (-1)^n$ which diverges and thus is not cauchy.

(c) Not cauchy. Consider $a_n = \frac{(-1)^n}{n}$. Since $a_n \to 0$, a_n is cauchy. But

$$c_n = \begin{cases} 0, & n \text{ even} \\ -1, & n \text{ odd} \end{cases}$$

Thus, the sequence is alternating and never converges. Therefore c_n is not cauchy.